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Unit 5 - Knowledge Check

PDF Documents

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UNIT 5 KNOWLEDGE CHECK

1 Why is alarm technology important for people with disabilities?

- ☐ A It helps you read faster.
- ☐ B Enables safe response to emergencies.
- ☐ C Helps you see better.
- ☐ D The alarm sound becomes louder.

CORRECT ANSWER

☒ B

2 Select all examples of alert technology used for deaf people.

- ☐ A Audio Alarm
- ☐ B Flashing light alarm
- ☐ C vibrating bell
- ☐ D clock alarm

CORRECT ANSWER

☒ B, ☒ C

3 When developing a visual warning system, which modules are unsuitable for safely simulating fire situations?

- ☐ A Display module
- ☐ B Speaker module
- ☐ C Dial module
- ☐ D Environment module

CORRECT ANSWER

☐ ?

4 When two people draw with 4 cards, A wins if an even number is drawn, and B wins if an odd number is drawn. When expressed in code so that A wins, which of the following commands is appropriate?

- ☐ A 2 and 4
- ☐ B 2 or 4
- ☐ C break
- ☐ D continue

CORRECT ANSWER

☒ B

5 What features should be added to LED alarms to indicate ongoing hazards such as fire?

- (A) Music must be playing.
- (B) It should blink and then stay on.
- (C) It should turn off automatically.
- (D) The color should change quickly.

CORRECT ANSWER

(D)

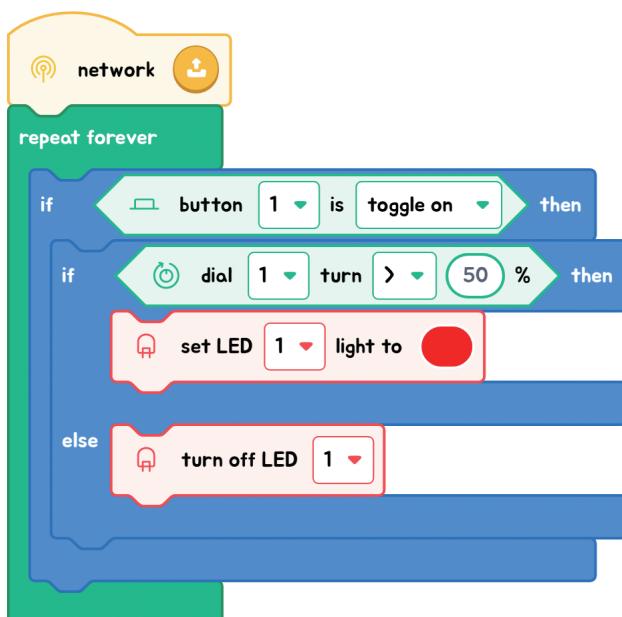
6 Why do programs use nested conditionals?

- (A) To make the code easier to read
- (B) To simplify simple operations
- (C) To handle complex decision-making processes through multiple stages of verification.
- (D) To add a loop within the condition

CORRECT ANSWER

(C)

7 Look at the following coding to determine the function.



- (A) Even if the button toggle is not turned on, the LED lights up red if the dial exceeds 50.
- (B) After the button toggle is turned on, the LED is turned on only when the dial is positioned above 50.
- (C) The LED turns off When the button toggle is turned off.
- (D) If the button toggle is on and the dial position is 50 or higher, the LED will turn on in any order.

CORRECT ANSWER

(D)



Critical Thinking Challenge

- 1 Design a new alarm system for a family member who is hearing impaired. What special features would you add to make the system effective and user-friendly? Think about how these features could help them respond quickly to emergencies and make their daily life safer.

- 2 A deaf student in your school often misses important announcements and emergency alarms. Your challenge is to create a plan for implementing an alert system throughout the school that ensures they are notified promptly. What strategies and technologies would you use, and how would you address potential issues such as device failures or distractions?

EXPANDING KNOWLEDGE ON THE TOPIC

**Multi-Sensory Alarm Systems for Inclusive Safety:**

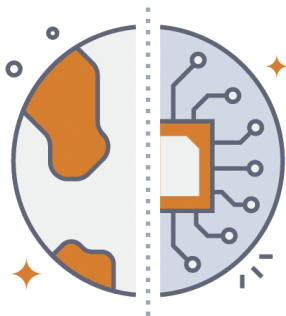
Multi-sensory alarm systems utilize a combination of visual, auditory, and tactile feedback to create comprehensive alert solutions for individuals with disabilities. These systems work by engaging multiple senses to ensure effective communication during emergencies. By incorporating flashing lights, vibrations, and other feedback mechanisms, multi-sensory alarms enhance awareness and response times, providing critical support for those who rely on non-auditory cues.

The Integration of Smart Technology in Personal Safety

Innovative technology, such as connected devices and IoT systems, significantly enhances the safety of deaf individuals. These technologies enable real-time alerts through connected home systems, integrating various sensors and devices that notify users via visual or tactile signals. By linking alarms, doorbells, and environmental sensors, smart homes create a network that ensures residents' continuous safety and peace of mind.

**Community-Wide Solutions for Accessibility in Safety**

Communities worldwide are adopting public alert systems tailored to include individuals with hearing impairments. Schools, workplaces, and public venues equip spaces with visual alarms, vibrating floor panels, and other inclusive technologies that warn during emergencies. Such systems are designed to create equitable environments where everyone can respond promptly and safely regardless of hearing ability.

The Advancements in Wearable Technology for Safety Alerts:

Wearable technology has emerged as a practical solution for ensuring safety among individuals with hearing impairments. Devices such as smartwatches and specially designed wristbands can vibrate or display visual notifications to alert users of emergencies or important events. These portable and customizable wearables are valuable tools for personal safety and convenience.

Challenges and Innovations in Emergency Response for the Hearing Impaired:

Hearing-impaired individuals face unique challenges during emergencies, especially when visual cues are limited, such as in the dark or while sleeping. Innovations in alert technology, like bed shakers, vibrating alarms, and intelligent lighting systems, address these challenges. These solutions ensure deaf people can receive alerts reliably, enhancing their safety and independence.